Introduction to probability theory

Computational modeling of language
Spring 2026

Bureaucracy

- **Discussion board**: Now in place on bCourses.

Bureaucracy

- **Discussion board**: Now in place on bCourses.
- **In case of illness**: Please stay home and get better - we will accommodate!

Last week

- The pumping lemma
- PDAs and context-free languages
- Chomsky vs. Shannon on human language
- On toward numbers
- Pandas, Karen, Terry

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- Introduction to probability theory
- Bayes' rule
- Marr's levels of analysis
- Markov chains

Small groups

Trade notes on **last week’s material**, and **this week’s readings**, to generate **comments** and/or **questions**.

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This week

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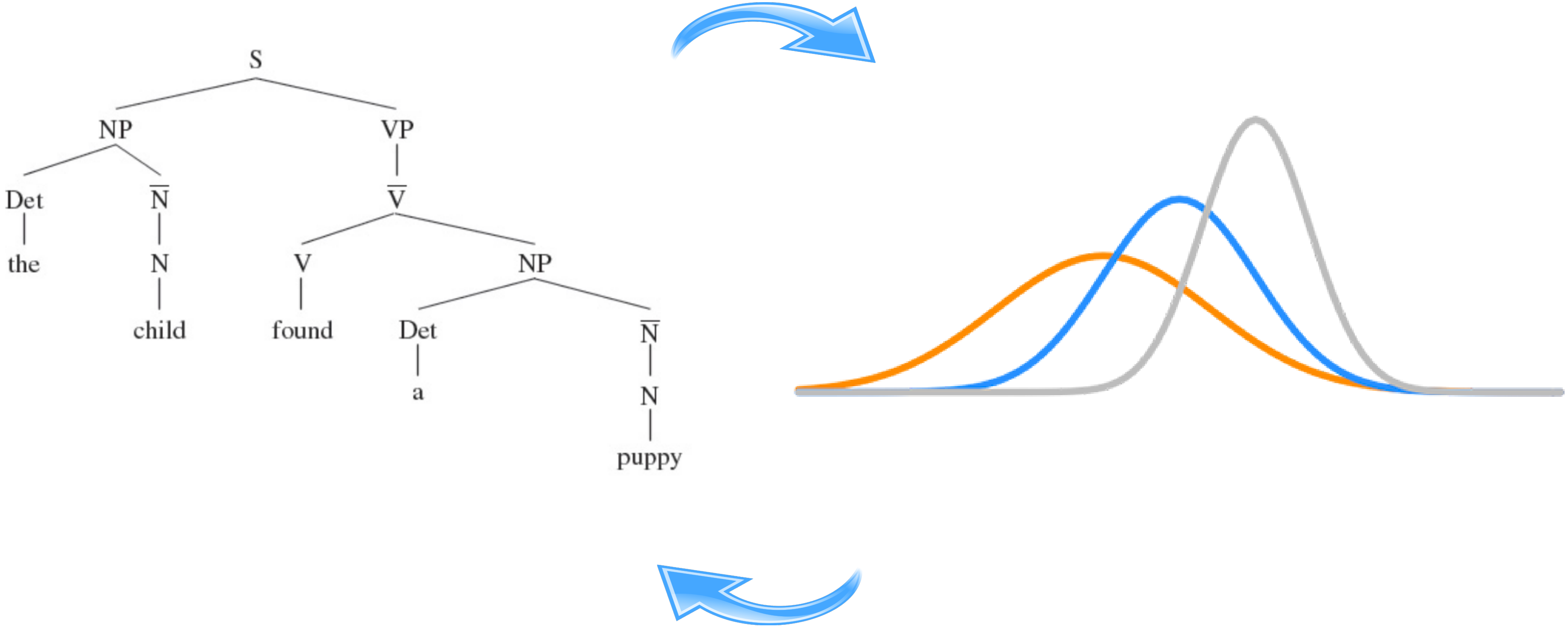
- Bayes’ rule
- Marr’s levels of analysis
- Markov chains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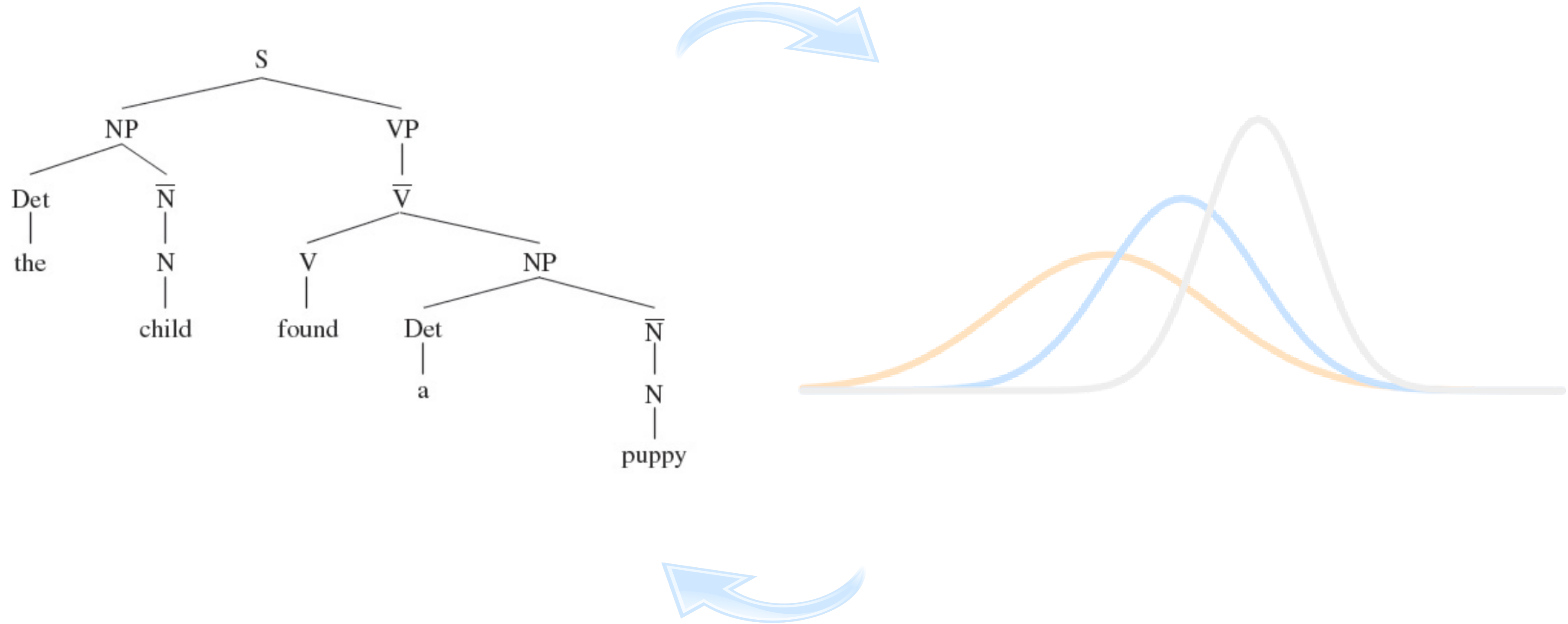
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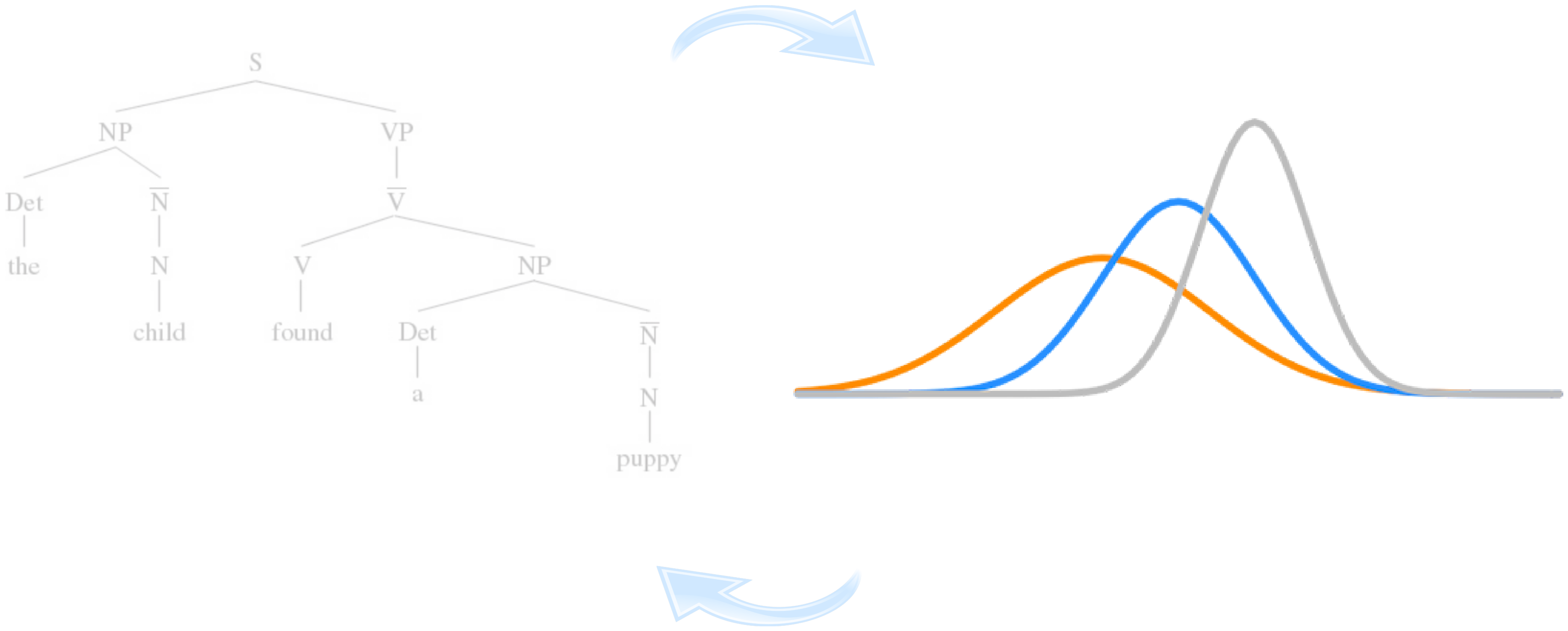
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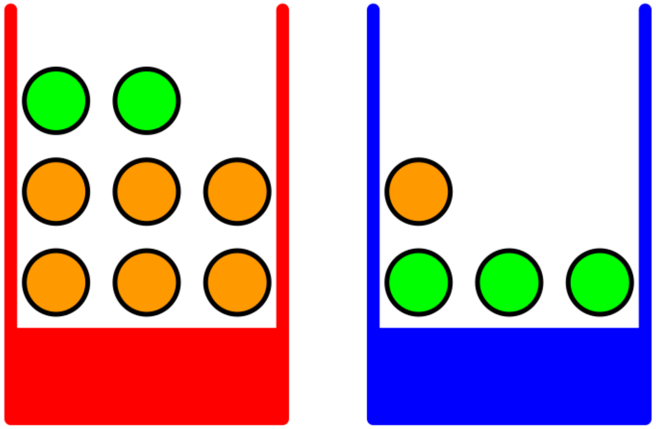
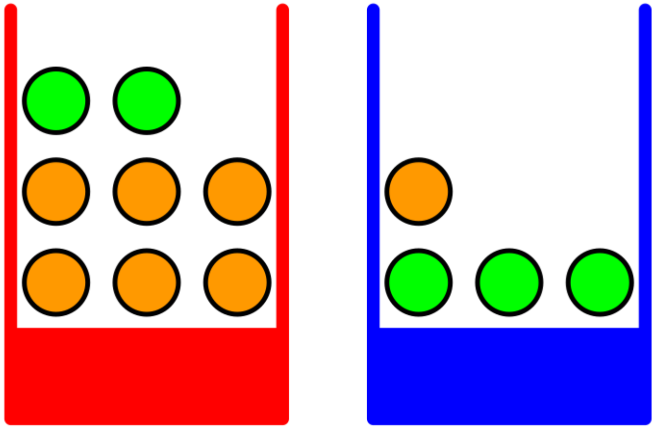
Probability theory

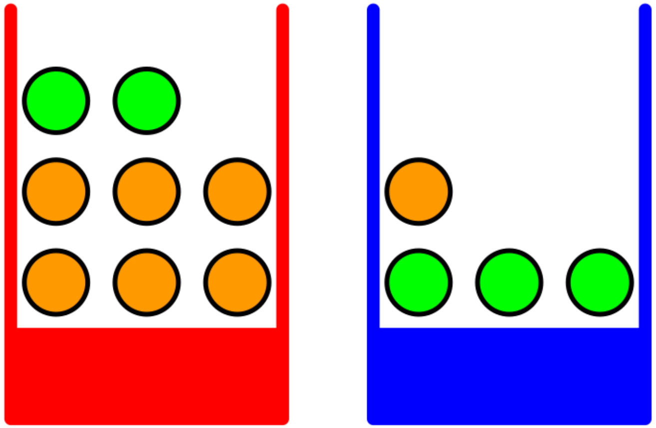
a consistent framework for the quantification and manipulation of **uncertainty**

Bishop 2006

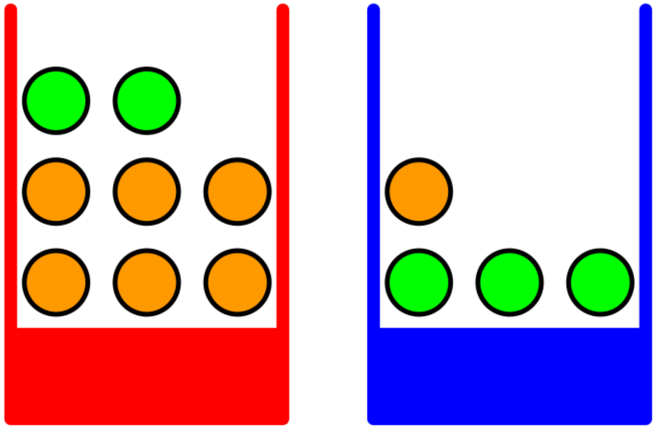




- Random variable: a variable whose possible values reflect outcomes of a random phenomenon.



- Random variable: a variable whose possible values reflect outcomes of a random phenomenon.
- E.g. B: the color of a selected box (r or b)



$p(B=r) = .4$
$p(B=b) = .6$

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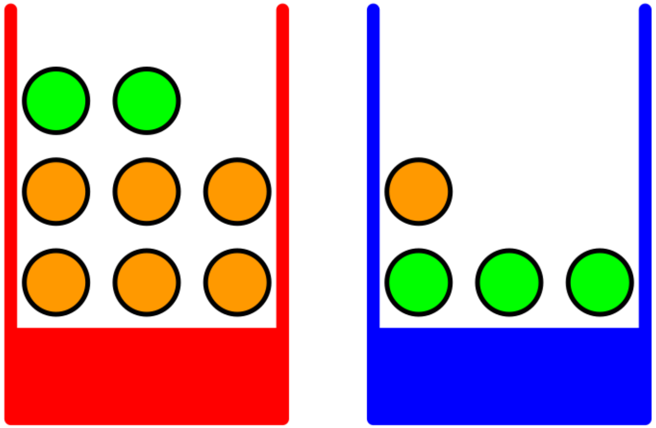

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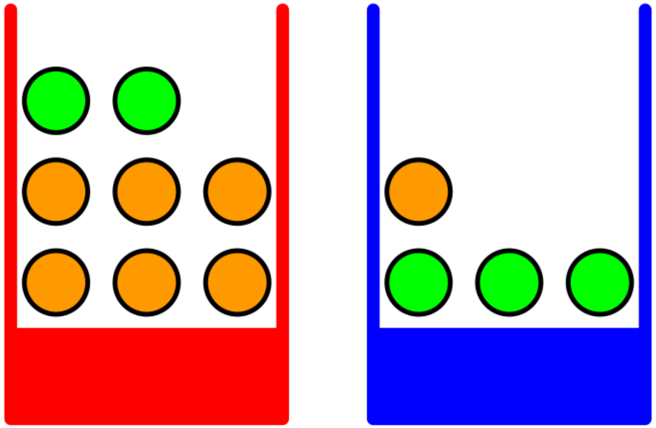
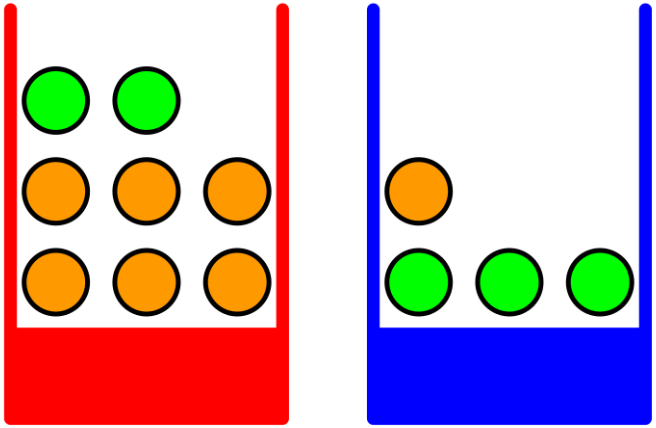


$p(B=r) = .4$
$p(B=b) = .6$

$p(F=a|B=r) =$
$p(F=o|B=r) =$

- <u>Random variable</u>: a variable whose possible values reflect outcomes of a random phenomenon.
- E.g. *B*: the color of a selected box (*r* or *b*)
- E.g. *F*: selected fruit (*a* or *o*)



$p(B=r) = .4$
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$p(F=a|B=r) = 2/8$
$p(F=o|B=r) = 6/8$

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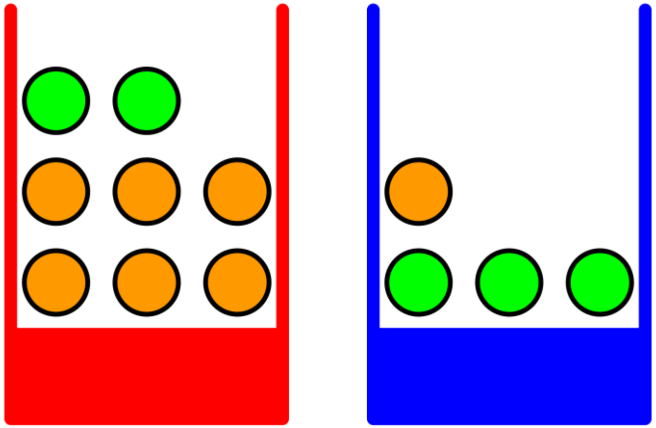
Interpretation of probability

Frequentist vs. **Bayesian**

Interpretation of probability

Frequentist: Probability measures **proportion of outcomes in repeated trials**.



$p(F\!=\!a|B\!=\!r) = 2/8$
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Interpretation of probability

Frequentist: Probability measures **proportion of outcomes in repeated trials**.

- $p(Weather{=}rain|Date{=}Feb\ 11)$ is (the limit of) the proportion of February 11s in the historical record during which it rained.
- $p(Weather{=}rain|Date{=}Feb\ 11\ \mathbf{2026})$ is strictly meaningless – or trivially either 0 or 1, depending on whether it rains tomorrow or not.

Interpretation of probability

Bayesian: Probability measures **degree of belief**.

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- Here, $p(\mathit{Weather}{=}\mathit{rain} \mid \mathit{Date}{=}\mathit{Feb}\ \mathit{11}\ \boldsymbol{2026})$ is meaningful.

Interpretation of probability

Bayesian: Probability measures **degree of belief**.

- Here, $p(\mathit{Weather}{=}\mathit{rain} | \mathit{Date}{=}\mathit{Feb\ 11}\ \boldsymbol{2026})$ is meaningful.
- It measures the **degree of belief** that it will rain, given that the date is Feb. 11 2026.

2 random variables: X, Y

- X can take on values x_1, x_2, x_3, …
- Y can take on values y_1, y_2, y_3, …

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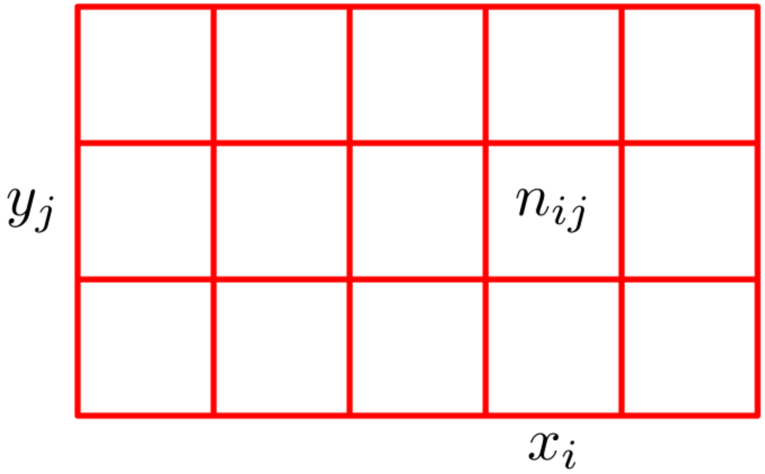
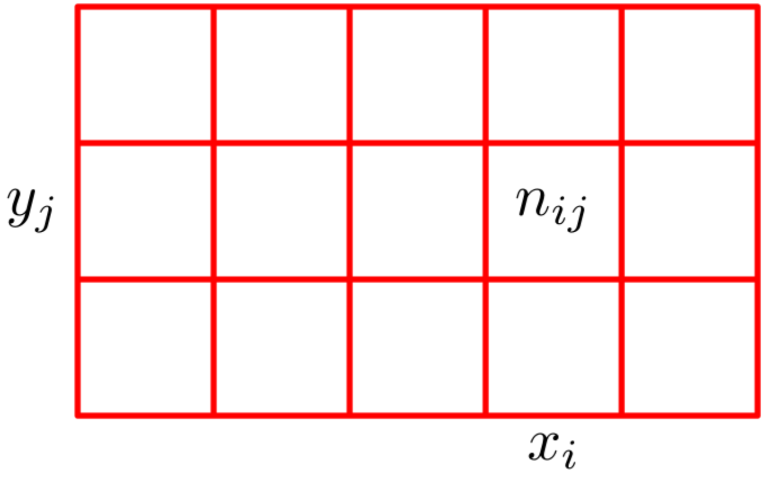
- Y can take on values y_1, y_2, y_3, …
- And we can **count how often** particular **pairs of values** x_i, y_j occur. That count is n_{ij}.

2 random variables: *X*, *Y*



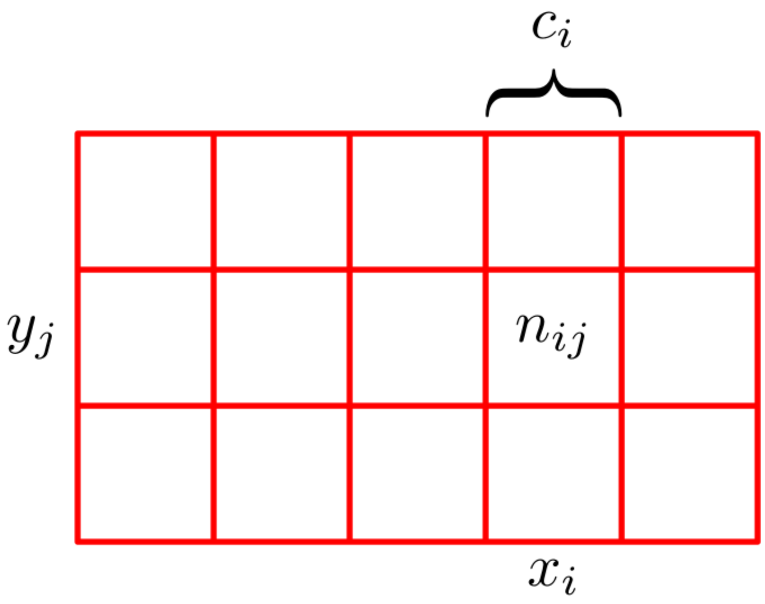
- *X* can take on values x_1, x_2, x_3, …
- *Y* can take on values y_1, y_2, y_3, …
- And we can **count how often** particular **pairs of values** x_i, y_j occur. That count is n_{ij}.

n_{ij} = count for x_i, y_j



n_{ij} = count for x_i, y_j

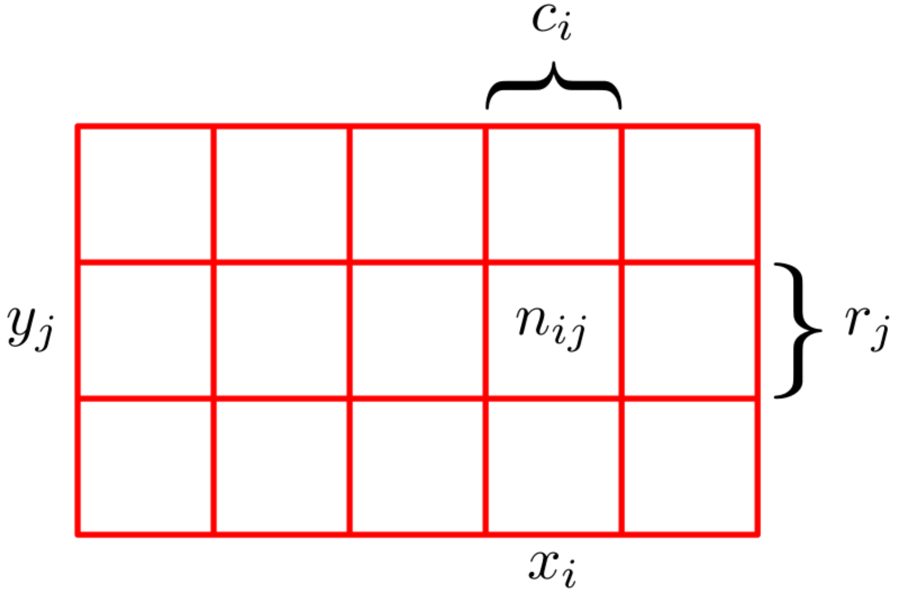
c_i = total count in column i



n_{ij} = count for x_i, y_j

c_i = total count in column i

r_j = total count in row j

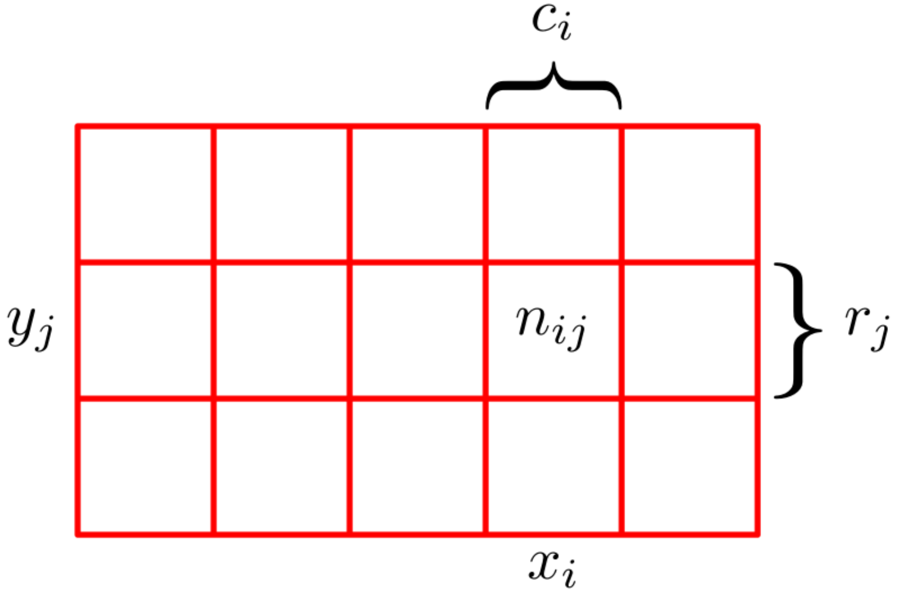


n_{ij} = count for x_i, y_j

c_i = total count in column i

r_j = total count in row j

N = total count in all cells



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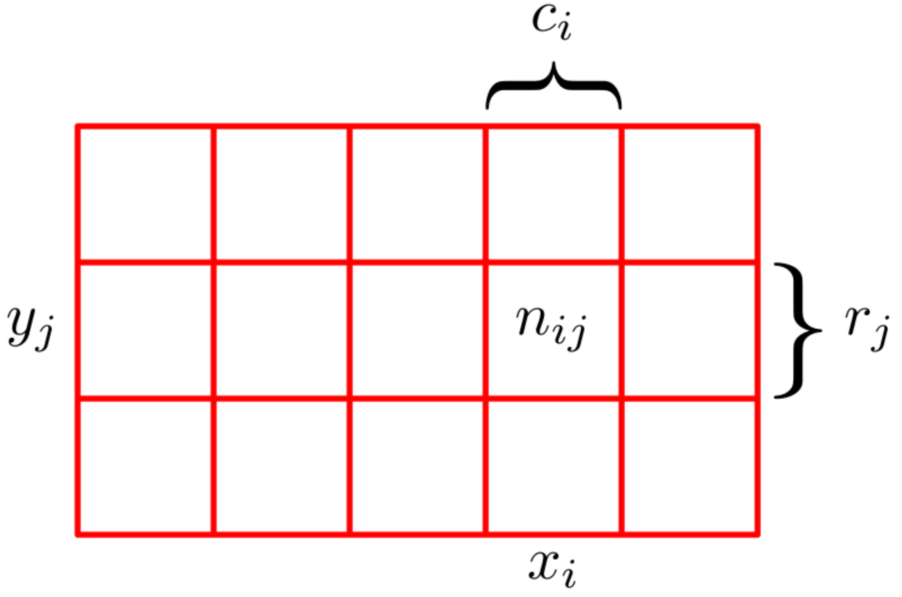

$p(X=x_i, Y=y_j) = ?$

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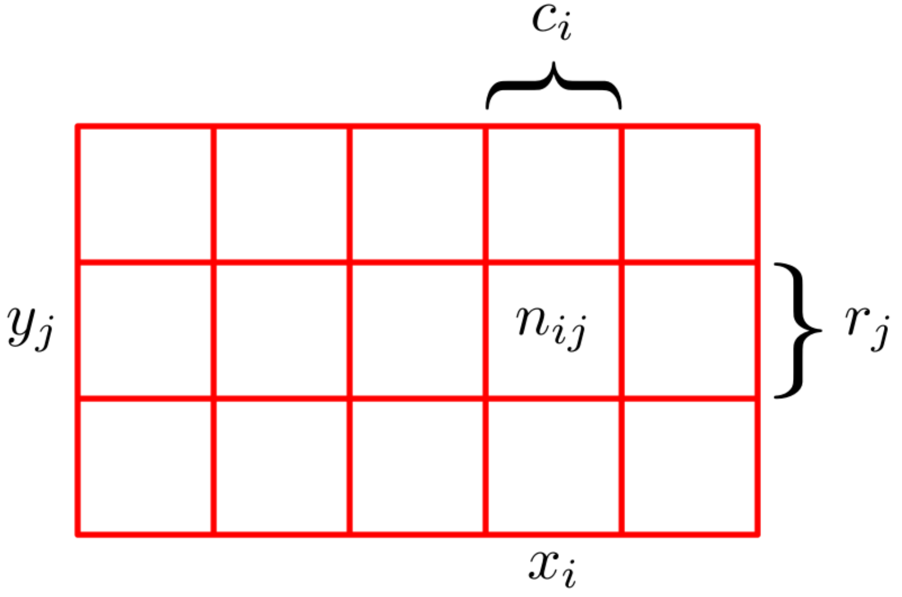


$p(X=x_i, Y=y_j) = n_{ij} / N$

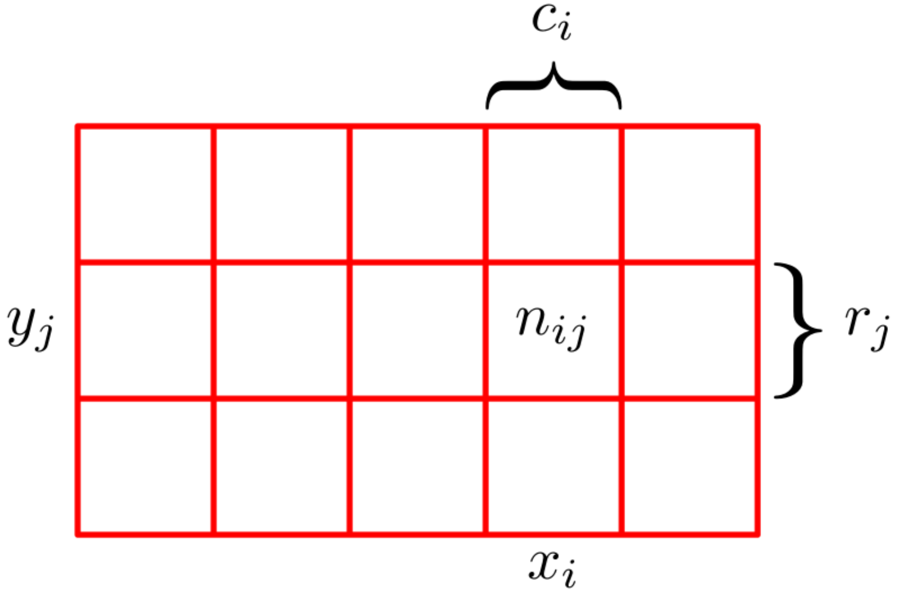
n_{ij} = count for x_i, y_j

c_i = total count in column i

r_j = total count in row j

N = total count in all cells

$p(X{=}x_i)$ = ?



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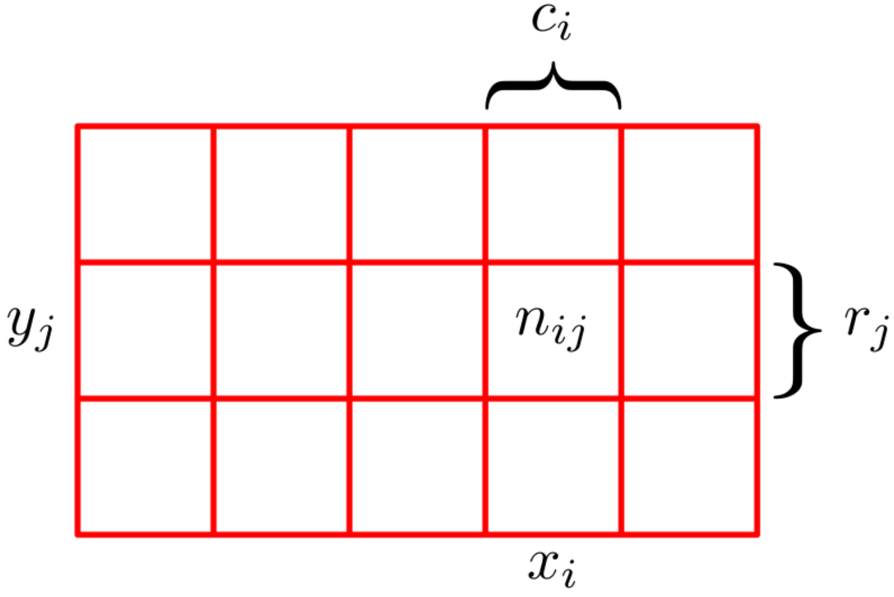

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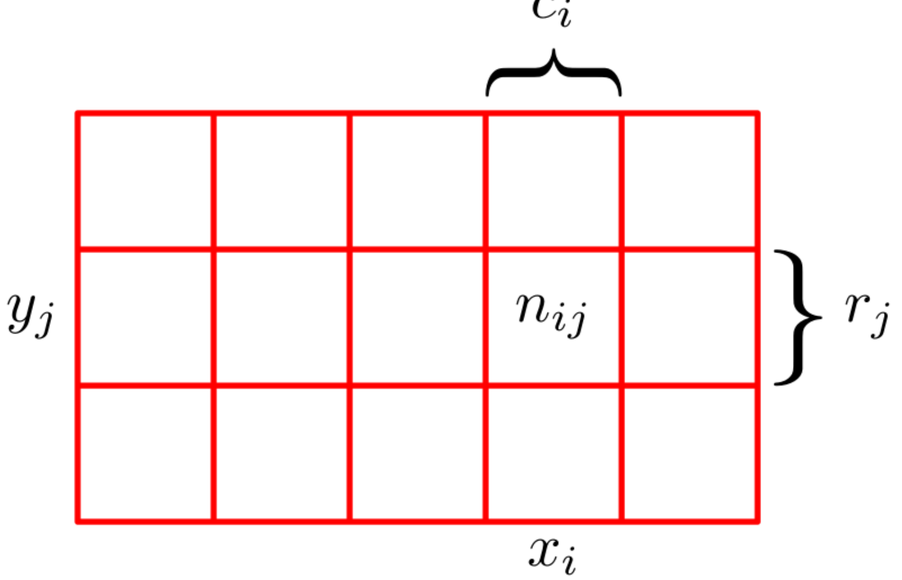

$$p(X=x_i) = c_i/N = (n_{i1} + n_{i2} + n_{i3}) / N$$

n_{ij} = count for x_i, y_j

c_i = total count in column i

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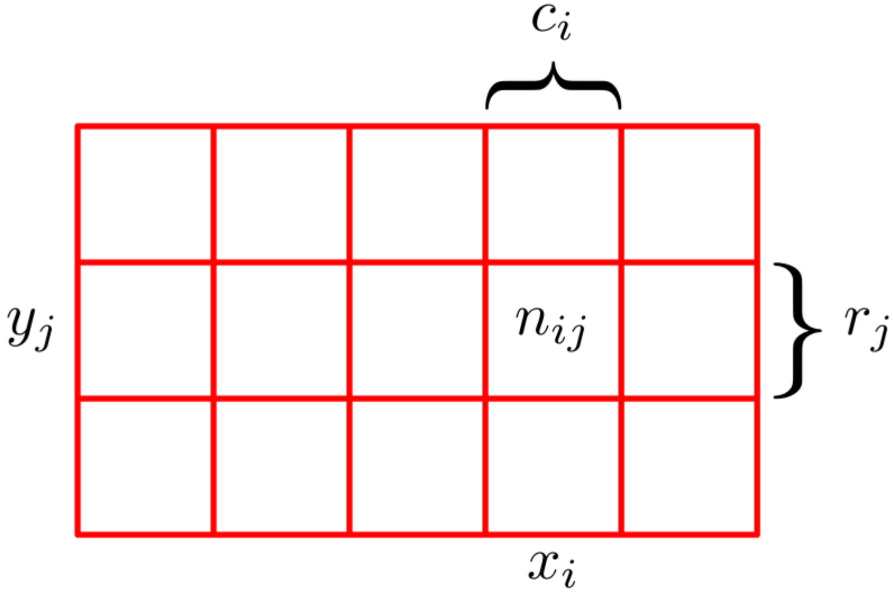

$$p(X=x_i) = c_i/N = (n_{i1} + n_{i2} + n_{i3}) / N$$

$$= (n_{i1}/N) + (n_{i2}/N) + (n_{i3}/N)$$

n_{ij} = count for x_i, y_j

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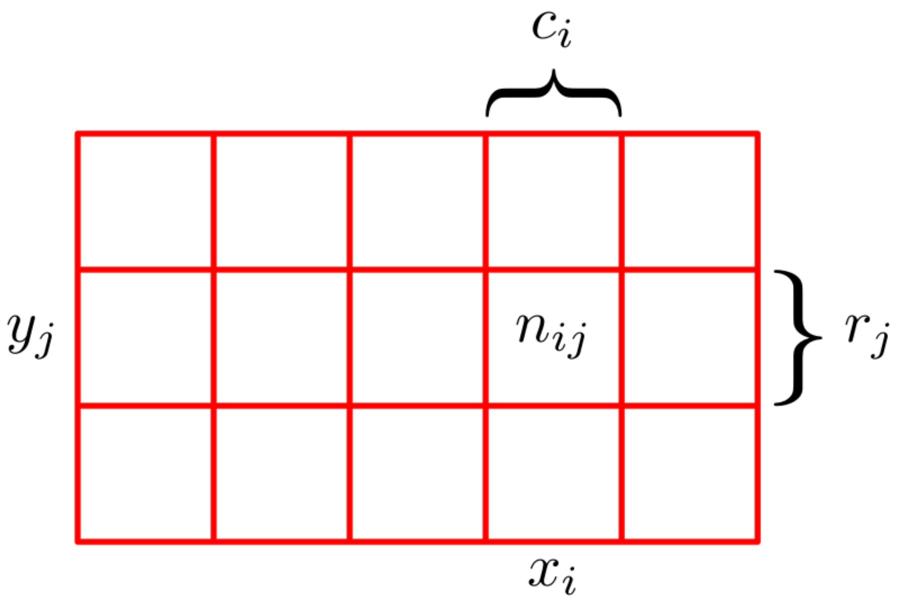

$$p(X=x_i) = c_i/N = (n_{i1} + n_{i2} + n_{i3}) / N$$
$$= (n_{i1}/N) + (n_{i2}/N) + (n_{i3}/N)$$
$$= p(X=x_i, Y=y_1) + p(X=x_i, Y=y_2) + p(X=x_i, Y=y_3)$$

n_{ij} = count for x_i, y_j

c_i = total count in column i

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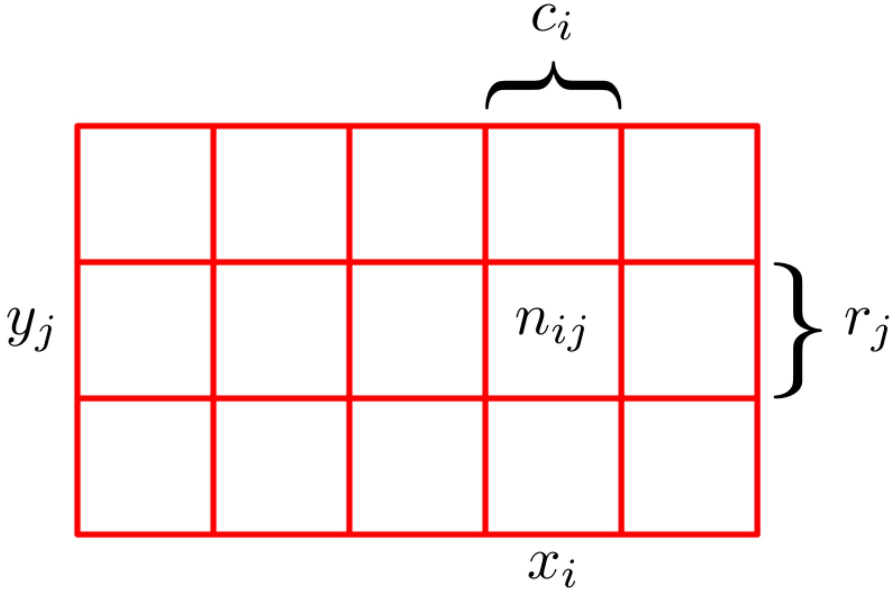


Sum rule: $$p(X = x_i) = \sum_{j=1}^{L} p(X = x_i, Y = y_j)$$

The Rules of Probability

sum rule $$p(X) = \sum_Y p(X, Y)$$

product rule $$p(X, Y) = p(Y|X)p(X).$$

Bayes' rule

$$p(A|B) = \frac{p(B|A) \quad p(A)}{p(B)}$$

Bayes' rule

posterior

$$\underline{p(A|B)} = \frac{p(B|A) \quad p(A)}{p(B)}$$

Bayes' rule

$$\underbrace{p(A|B)}_{\text{posterior}} = \frac{\underbrace{p(B|A)}_{\text{likelihood}} \; \underbrace{p(A)}_{\text{prior}}}{p(B)}$$

Bayes' rule

$$\overset{\text{posterior}}{p(A|B)} = \frac{\overset{\text{likelihood}}{p(B|A)}\;\overset{\text{prior}}{p(A)}}{\underset{\text{normalizing constant}}{p(B)}}$$

Bayes' rule

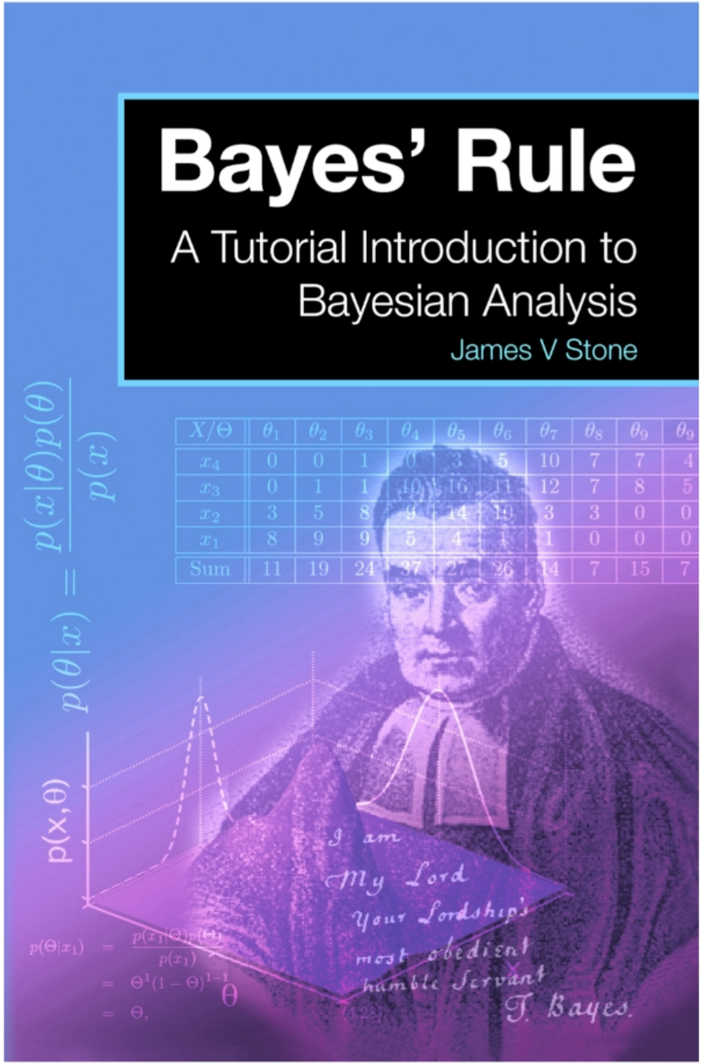
$$\underbrace{p(A|B)}_{\text{posterior}} = \frac{\overbrace{p(B|A)}^{\text{likelihood}}\ \overbrace{p(A)}^{\text{prior}}}{\underbrace{p(B)}_{\text{normalizing constant}}}$$

$$\underbrace{p(A|B)}_{\text{posterior}} \propto \underbrace{p(B|A)}_{\text{likelihood}}\ \underbrace{p(A)}_{\text{prior}}$$

Bayes' rule

$$\underbrace{p(H|d)}_{\text{posterior}} \propto \underbrace{p(d|H)}_{\text{likelihood}} \; \underbrace{p(H)}_{\text{prior}}$$

- *H*: hypothesis
- *d*: data



J. V. Stone 2013

It is remarkable that a science which began with the consideration of games of chance should have become the most important object of human knowledge.
Pierre Simon Laplace, 1812.

"... we balance probabilities and choose the most likely. It is the scientific use of the imagination ... "

Sherlock Holmes, The Hound of the Baskervilles.

AC Doyle, 1901.

?

"... we balance probabilities and choose the most likely. It is the scientific use of the imagination ... "

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J. V. Stone 2013

(a) Bayes

(b) Laplace

Figure 1.1.: The fathers of Bayes' rule. a) Thomas Bayes (c. 1701-1761). b) Pierre-Simon Laplace (1749-1827).

J. V. Stone 2013

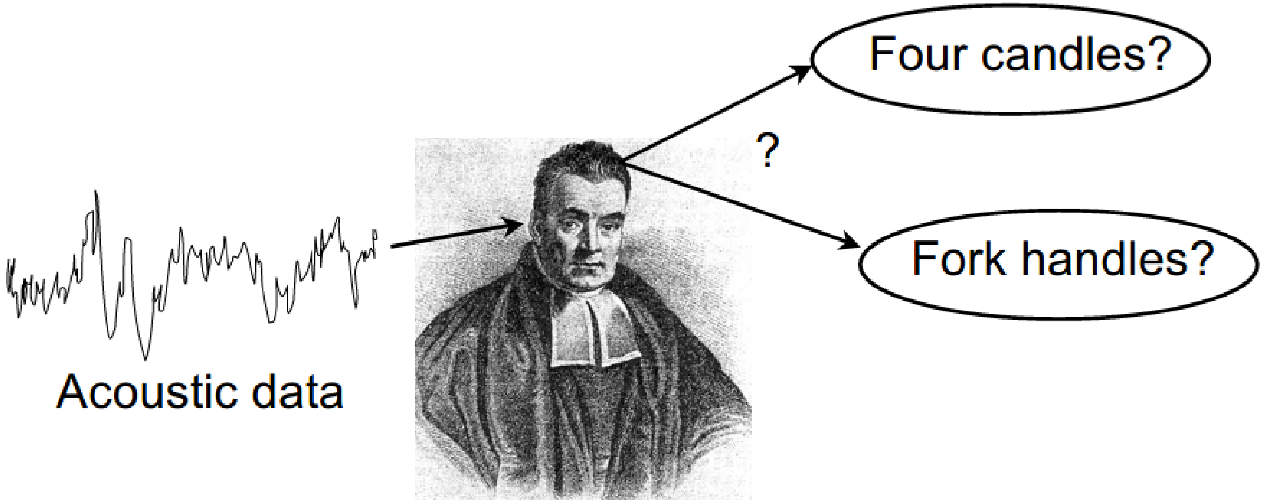
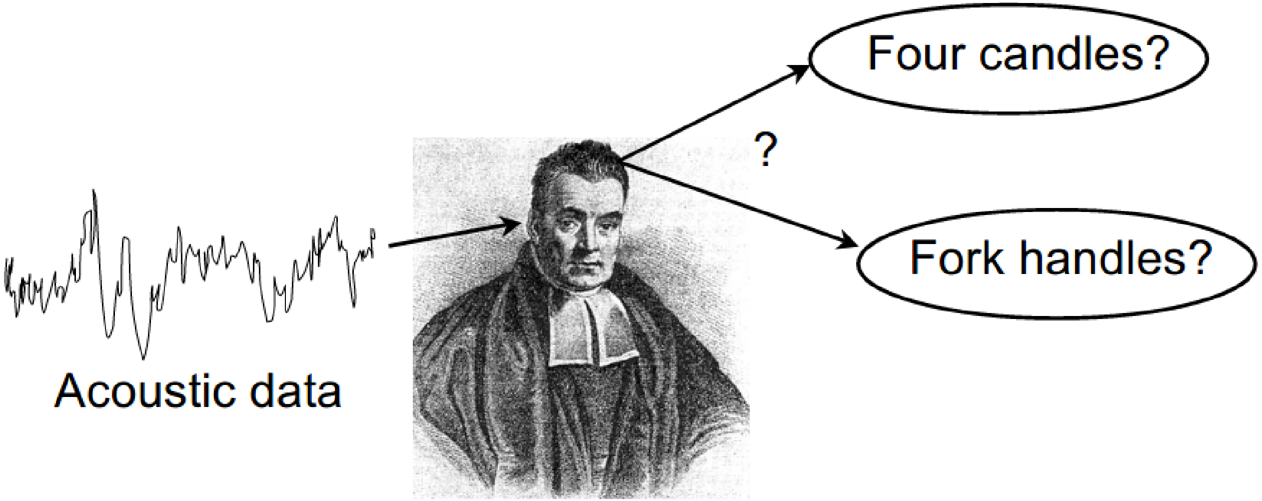
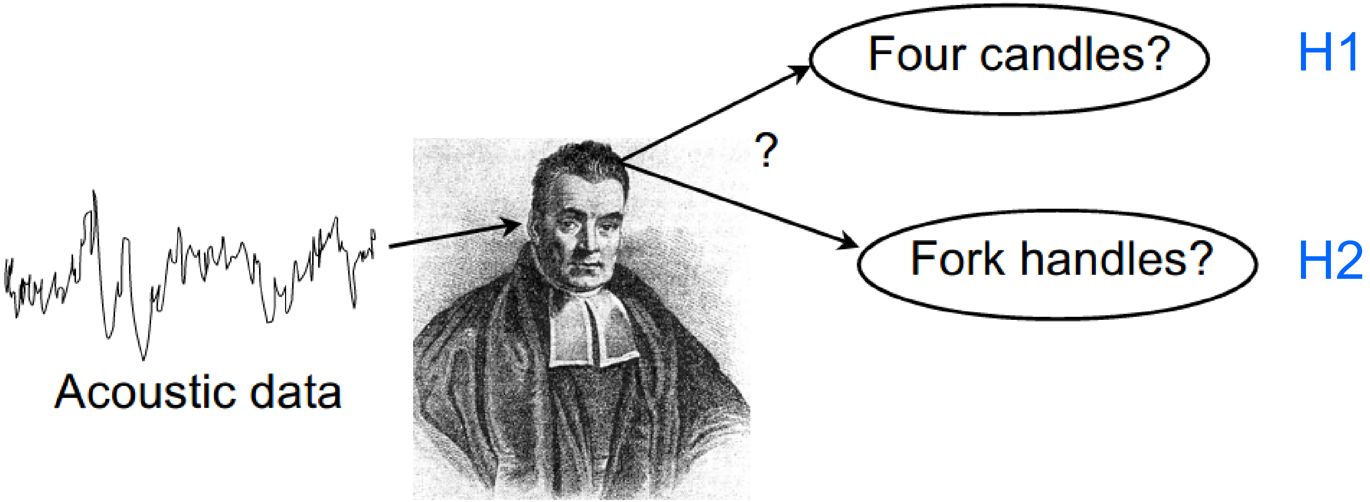


Figure 1.5.: Thomas Bayes trying to make sense of a London accent, which removes the *h* sound from the word *handle*, so the phrase *fork handles* is pronounced *fork 'andles*, and therefore sounds like *four candles* (see Fork Handles YouTube clip by The Two Ronnies).

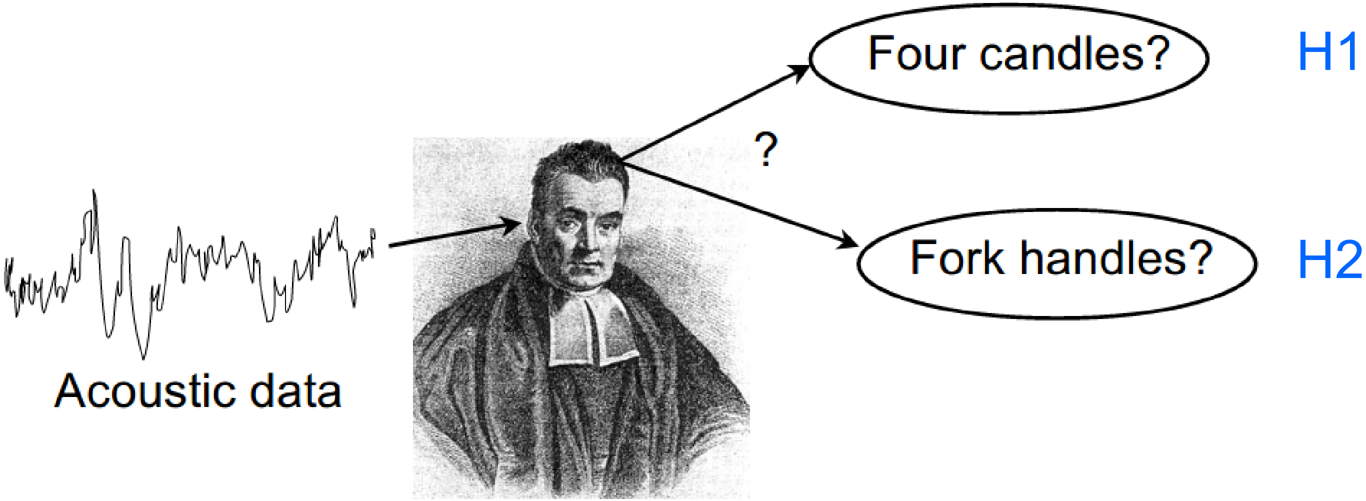
J. V. Stone 2013



$$\underbrace{p(H|d)}_{\text{posterior}} \propto \underbrace{p(d|H)}_{\text{likelihood}}\ \underbrace{p(H)}_{\text{prior}}$$



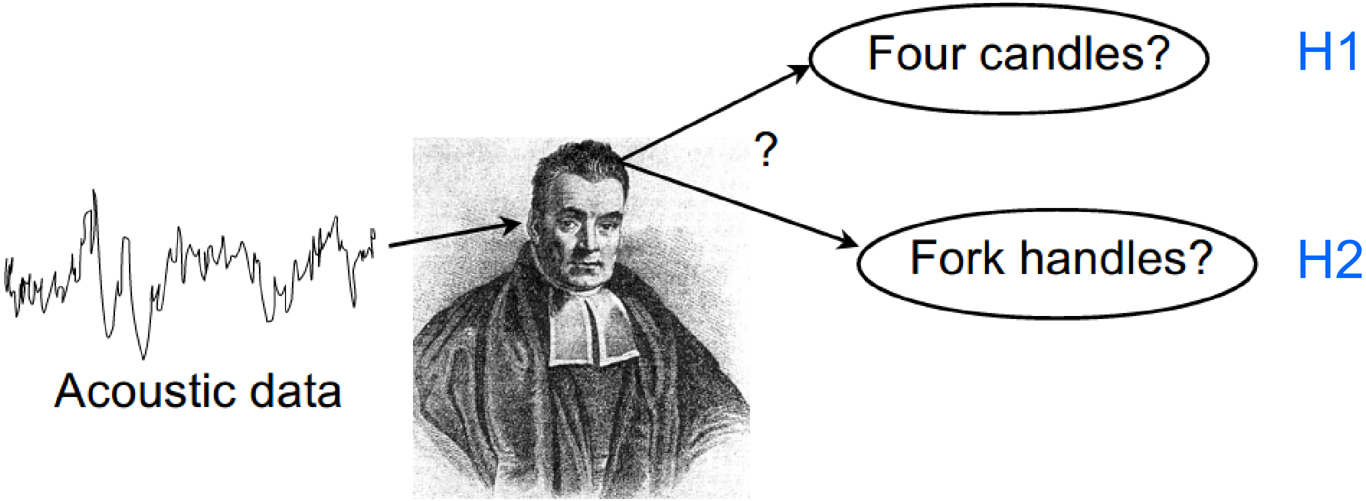
$$\overset{\text{posterior}}{\overline{p(H|d)}} \propto \overset{\text{likelihood}}{\overline{p(d|H)}}\ \overset{\text{prior}}{\overline{p(H)}}$$



Assume:

$p(d|H1) = p(d|H2)$

$$\underbrace{p(H|d)}_{\text{posterior}} \propto \underbrace{p(d|H)}_{\text{likelihood}} \; \underbrace{p(H)}_{\text{prior}}$$



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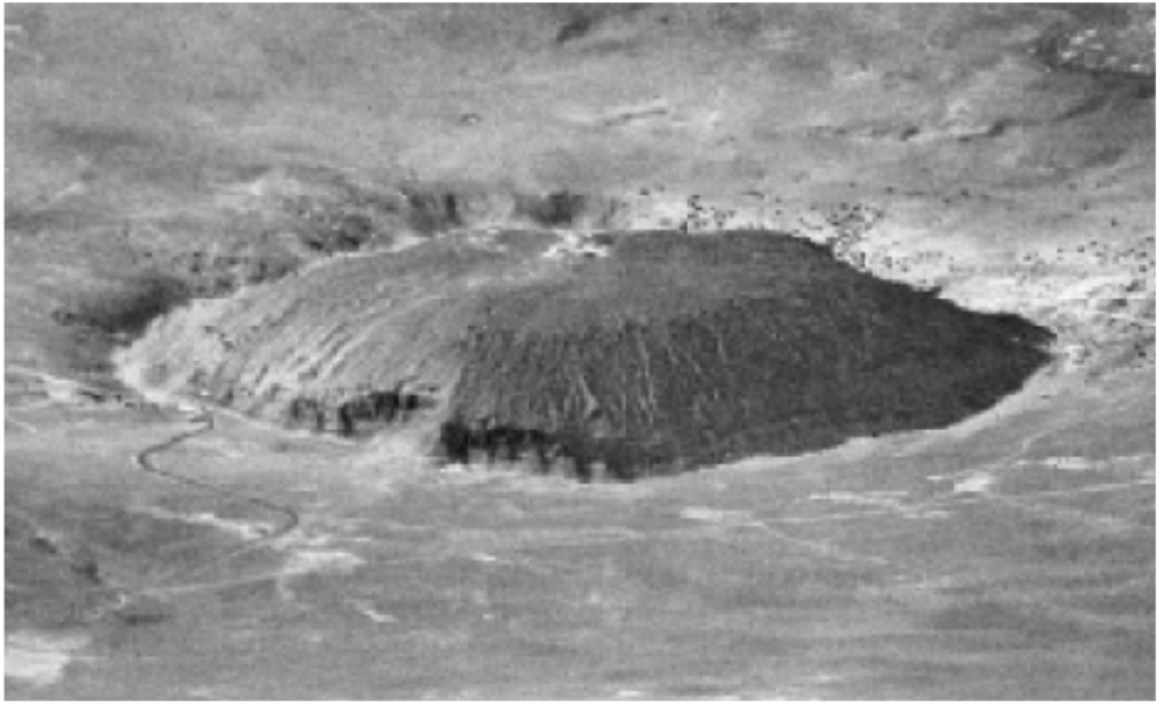
$$\underbrace{p(H|d)}_{\text{posterior}} \propto \underbrace{p(d|H)}_{\text{likelihood}}\ \underbrace{p(H)}_{\text{prior}}$$

Uttered in a candle store:

$p(H1) = .9$; $p(H2) = .1$



Seeing is not a direct apprehension of reality, as we often like to pretend. Quite the contrary: **seeing is inference from incomplete information**...
E.T. Jaynes 2003

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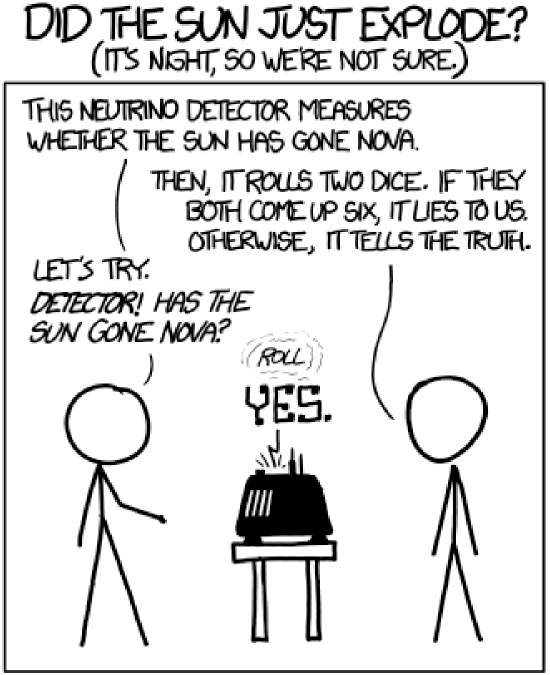
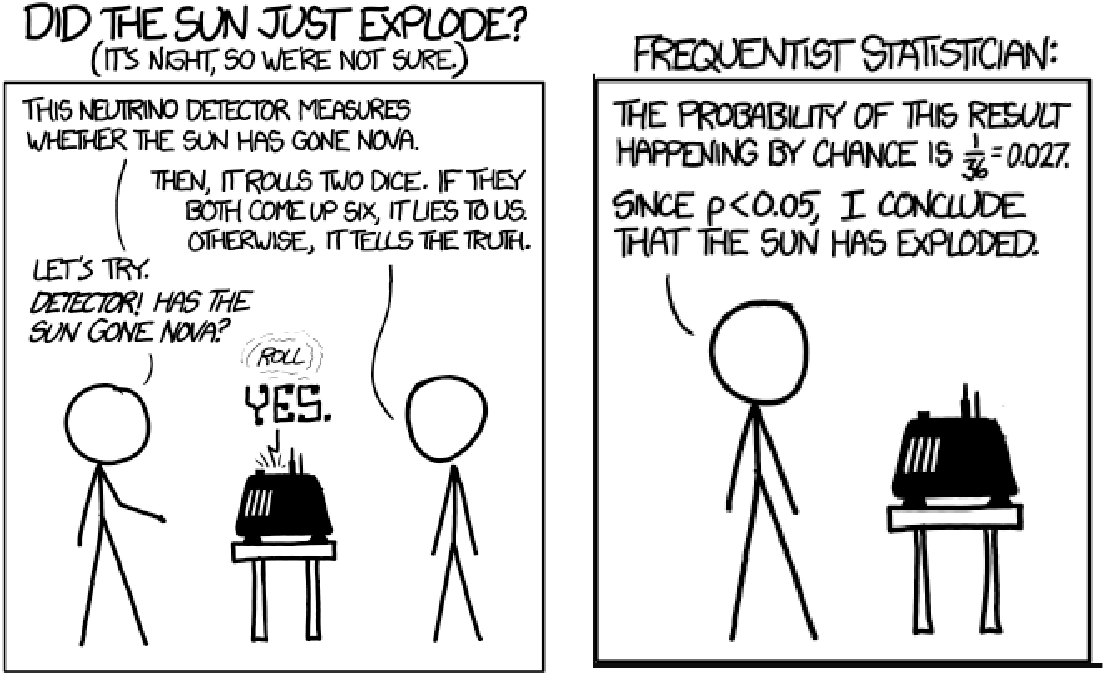
See also Helmholtz on
perception as unconscious inference.


DID THE SUN JUST EXPLODE?
(IT'S NIGHT, SO WE'RE NOT SURE.)
THIS NEUTRINO DETECTOR MEASURES WHETHER THE SUN HAS GONE NOVA.
THEN, IT ROLLS TWO DICE. IF THEY BOTH COME UP SIX, IT LIES TO US. OTHERWISE, IT TELLS THE TRUTH.
LET'S TRY.
DETECTOR! HAS THE SUN GONE NOVA?
(ROLL)
YES.


DID THE SUN JUST EXPLODE?
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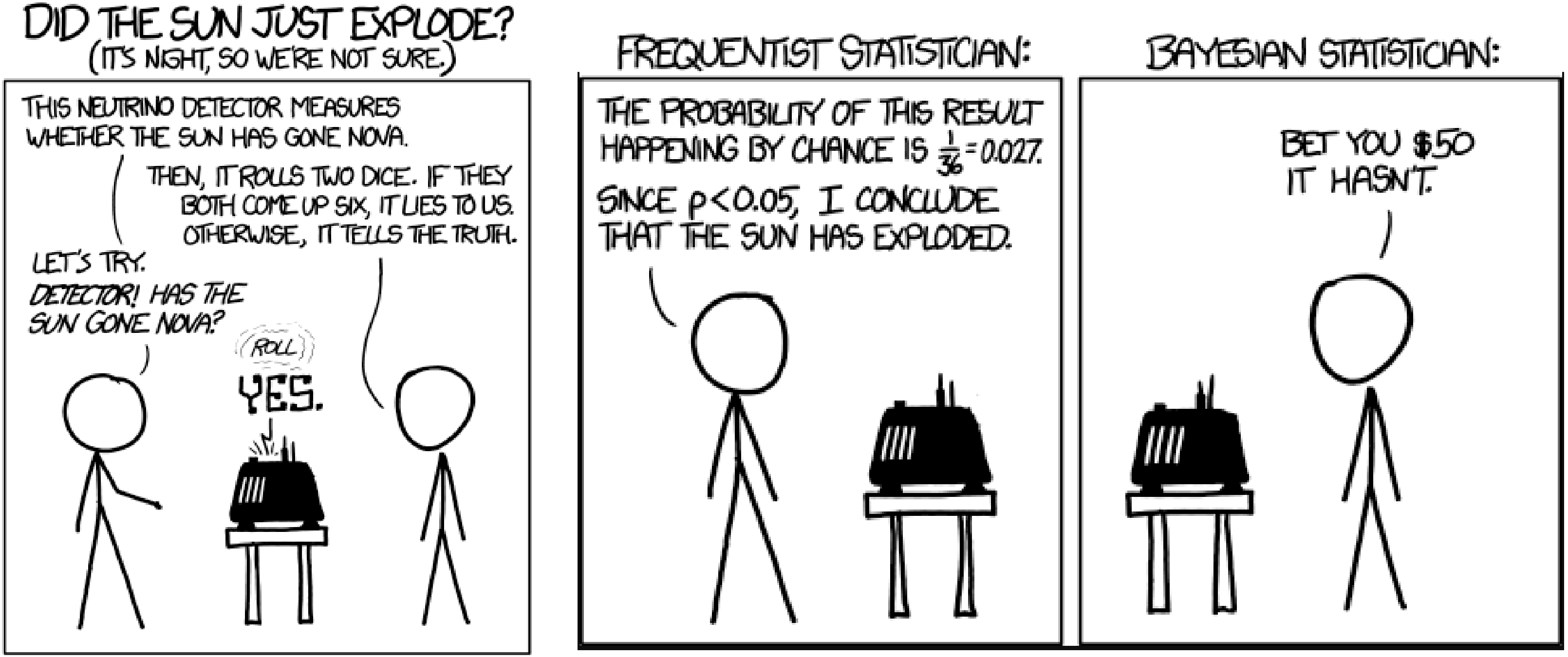
THEN, IT ROLLS TWO DICE. IF THEY BOTH COME UP SIX, IT LIES TO US. OTHERWISE, IT TELLS THE TRUTH.
LET'S TRY.
DETECTOR! HAS THE SUN GONE NOVA?
(ROLL)
YES.
FREQUENTIST STATISTICIAN:
THE PROBABILITY OF THIS RESULT HAPPENING BY CHANCE IS $\frac{1}{36}$ = 0.027.
SINCE $p < 0.05$, I CONCLUDE THAT THE SUN HAS EXPLODED.



DID THE SUN JUST EXPLODE?
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THIS NEUTRINO DETECTOR MEASURES WHETHER THE SUN HAS GONE NOVA.
THEN, IT ROLLS TWO DICE. IF THEY BOTH COME UP SIX, IT LIES TO US. OTHERWISE, IT TELLS THE TRUTH.
LET'S TRY.
DETECTOR! HAS THE SUN GONE NOVA?
ROLL
YES.
FREQUENTIST STATISTICIAN:
THE PROBABILITY OF THIS RESULT HAPPENING BY CHANCE IS 1/36 = 0.027.
SINCE p < 0.05, I CONCLUDE THAT THE SUN HAS EXPLODED.
BAYESIAN STATISTICIAN:
BET YOU $50 IT HASN'T.

Probabilistic inference, using the ideas we have seen here, is often taken as a **normative standard** against which human performance can be compared.

Marr’s (1982) levels of analysis

The **Marr Prize**, named in honor of David Marr, is given annually to the **best student paper** at the Cognitive Science Society conference.

cognitivesciencesociety.org

Marr’s three levels

The three levels at which any machine carrying out an information-processing task must be understood

Computational theory	*Representation and algorithm*	*Hardware implementation*

Marr's three levels

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Computational theory	*Representation and algorithm*	*Hardware implementation*
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?		

Marr's three levels

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Computational theory	*Representation and algorithm*	*Hardware implementation*
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	

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Computational theory	*Representation and algorithm*	*Hardware implementation*
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	How can the representation and algorithm be realized physically?

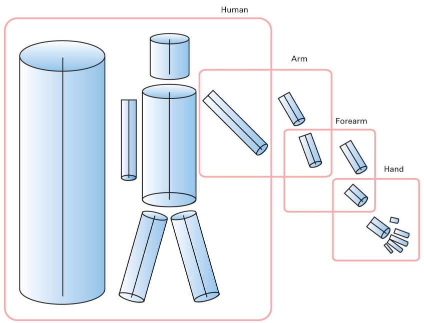
Marr's three levels

The three levels at which any machine carrying out an information-processing task must be understood

Computational theory	*Representation and algorithm*	*Hardware implementation*
The functional goal of the visual system is to determine the shape of objects in the world.		

Marr’s three levels

The three levels at which any machine carrying out an information-processing task must be understood

Computational theory	*Representation and algorithm*	*Hardware implementation*
The functional goal of the visual system is to determine the shape of objects in the world.		

Marr's three levels

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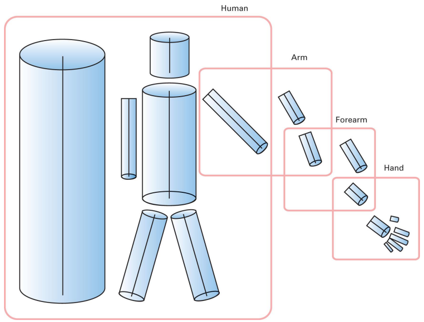
The functional goal of the visual system is to determine the shape of objects in the world.		Neurons, somehow.

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What about language?

The computational level

Specifies “what the system is up to” – what the **functional goal** of its processing is.

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And do humans do it that way – regardless of algorithmic and implementational detail?

The computational level

Marr: “trying to understand perception by studying only neurons is like trying to understand bird flight by studying only feathers. It just cannot be done.”

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This concerns Marr’s **computational** level.